Grace Feng

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Education

University of Michigan — Ann Arbor, MI

Bachelor of Science in Computer Science Engineering, GPA: 3.981/4.00 — *August 2024 – May 2027*

- Honors/Awards: Victor's Award Grant, National Merit Siemens Scholarship, William J. Branstrom Freshman Prize, University Honors, Class of 1931E Scholarship Finalist
- Activities: Peer Mentor for ENGR 100 - Underwater Vehicle, Project RISHI, Career Fair Website Committee
- Coursework: Statistics, Machine Learning, Operating Systems, Computer Security, Computer Organization, Cryptography, Data Structures & Algorithms, Linear Algebra, Automated Programming

Experience

Software Team — Ann Arbor, MI

Michigan Underwater Robotic Submarine — *August 2025 – Present*

- Migrated the autonomous underwater vehicle codebase from ROS1 to ROS2, ensuring continued framework support and improving long-term maintainability for the team's yearly competition cycle
- Built a real-time object classification model using YOLOv5 and YOLOv8 to identify underwater obstacles, feeding results into dead-reckoning navigation decisions during competition runs

Machine Learning Intern — Ann Arbor, MI

Nuclear Engineering and Radiological Sciences — *May 2026 – Present*

- Benchmarked two neural network pipelines for classifying raw waveform pulses as gamma or neutron events, finding one to be 7200x more computationally efficient
- Trained models in TensorFlow, running experiments on the University of Michigan Great Lakes computing cluster

Software Development Intern — Oak Ridge, TN

Research Student Internship, Oak Ridge National Laboratory — *May 2025 – August 2025; May 2026 – 2026*

- Built a custom data visualization tool in Python to transform and visualize raw phenotypic data for GWAS analysis on the DOE Knowledgebase (KBase) platform (2025)
- Upgraded and updated four bioinformatics tools (HISAT2, Bowtie2, DESeq2, CheckM2) within KBase to align with the latest third-party versions (2026)
- Constructed user-facing tutorials for RNA-seq and GWAS analysis workflows
- Gained experience applying analysis pipelines in computational biology, including genome annotation, GTDB-Tk, metabolic modeling, and sequence analysis

Projects

Mendelian Randomization of COVID and AKI — Ann Arbor, MI

Zhou Research Lab — *October 2024 – 2026*

- Developed a Mendelian Randomization pipeline in R using two-sample MR methods to analyze the causal effects of COVID-19 on Acute Kidney Injury
- Presented findings at the National Conference of Undergraduate Research

Patient Prediction Model

EECS 445- Machine Learning

- Developed a prediction model for patient mortality using the scikit-learn Python package
- Selected features and tuned hyperparameters for Logistic Regression and Radial Basis Vector Kernel
- Achieved a 100% increase in F1 score measure compared to baseline approach, indicating robustness of feature engineering and filtering methods

Websketcher

EECS 498- Automated Programming

- Developed a platform to construct HTML from hand-drawings and natural language instructions
- Integrated OpenCV computer vision model for image parsing with ¡5% on all benchmarks

Honors & Awards

2024 NCWIT Aspirations in Computing Regional Winner; 2024 National Merit Finalist; 2023 USA Biology Olympiad Semifinalist (top 10% of test takers nationwide); 2025 University of Michigan Biostatistics Hackathon recognized project; 100 hours of volunteering for the Gear Up for STEM program

Technical Skills

Languages: Python, SQL, C++, C#, JavaScript, HTML/CSS, MATLAB, R
Tools & Platforms: Docker, ROS2, Tensorflow, Pytorch
Interests: Computer Science, Engineering, Computational Biology, Bioengineering, Robotics, Math